

## CCE SAFETY MANUAL

Contact: Mr. Jinu K.T. (956 74 20 776) Mr. Reynold Jose (994639411)

& Dr. Arun Augustin (9740641930)

PRINCIPAL: Mr. Sijo MT (9847520278)

College Front Office : 04802825355

FIRE & RESCUE Irinjalakkuda: 4802820558)

POLICE Irinjalakkuda: 0480 282 5228

Govt. Gen HOSPITAL:0480282 5213

May 2, 2026



**Principal**  
**Christ College of Engineering**  
**(Autonomous)**  
**Irinjalakuda -680 125**



# Contents

|          |                                                                                                                                 |          |
|----------|---------------------------------------------------------------------------------------------------------------------------------|----------|
| <b>1</b> | <b>EMERGENCY NUMBERS</b>                                                                                                        | <b>1</b> |
| <b>2</b> | <b>General Lab Safety and Security Procedures</b>                                                                               | <b>2</b> |
| 2.1      | Equipment and Facility Operating Procedures . . . . .                                                                           | 2        |
| 2.2      | Common sense traditions . . . . .                                                                                               | 3        |
| 2.3      | EMERGENCY NUMBERS: MEDICAL . . . . .                                                                                            | 3        |
| 2.4      | Tools and workplace discipline . . . . .                                                                                        | 4        |
| 2.5      | Team coordination . . . . .                                                                                                     | 5        |
| 2.6      | Handling external requests/ cleaning / maintenance . . . . .                                                                    | 5        |
| 2.7      | Working with the Workshop: CAD; intelligent design; off-the-shelf vs. part fabrication; scheduling; priorities; charge numbers. | 6        |
| <b>3</b> | <b>Specifics Related to UAVs</b>                                                                                                | <b>7</b> |
| <b>4</b> | <b>Chemical Safety</b>                                                                                                          | <b>9</b> |
| 4.1      | MSDS: Material Safety Data Sheets . . . . .                                                                                     | 9        |
| 4.2      | NFPA Hazard Ranking System . . . . .                                                                                            | 10       |
| 4.3      | Procedure: Chemical Spills . . . . .                                                                                            | 10       |
| 4.3.1    | Chemical Spills Indoors . . . . .                                                                                               | 11       |
| 4.3.2    | Chemical Spills Outdoors . . . . .                                                                                              | 11       |

## CONTENTS

ii

|     |                               |    |
|-----|-------------------------------|----|
| 4.4 | Eye Protection . . . . .      | 11 |
| 4.5 | Wearing Gloves . . . . .      | 12 |
| 4.6 | Working Alone . . . . .       | 13 |
| 4.7 | GAS CYLINDERS . . . . .       | 13 |
| 4.8 | Machine tool safety . . . . . | 17 |
| 4.9 | BIBLIOGRAPHY . . . . .        | 18 |

## Chapter 1

# EMERGENCY NUMBERS

Remember, the most important thing is to get immediate aid for the injured, and clear everyone out to safety!!!

1. Apply first aid as necessary.
2. College Office 0480 282 5355 or 830 194 5355. Principal: 98475 20278
3. Police/emergency : 0480 282 5228
4. Irinjalakkuda Govt. General Hospital: 0480 282 5213

## Chapter 2

# General Lab Safety and Security Procedures

Memorize, understand, and absorb these. The detailed Rules are to be read so that you can see what dangers have been considered and why rules have been developed from hard experience.

We do our work at the Christ College of Engineering, Keralam, India. In all that is done here, we must observe all of the constraints / motivators below, generally equal in importance, but if it ever comes to a conflict, in descending order of priority:

1. Safety
2. The Law
3. The rules of the College
4. The practices of the specific department/project
5. The need to get work done, and make progress.
6. The need to be nice and pleasant to everyone.

### 2.1 Equipment and Facility Operating Procedures

Those are **\*\*NOT\*\*** included in this Manual because they are instructed/conveyed separately to those who are allowed to operate

them.

## 2.2 Common sense traditions

1. Team members come from all strata of life, with many different beliefs and experiences, but a common desire to study and do well in the things which we do. We focus on the things where we can work together and strive for excellence.
2. Everyone has their own beliefs and preferences and is generally welcome to them, consistent with the above priorities. We all read news media and are intelligent and thoughtful people. We do not discuss religion, politics, sex or personal wealth in the workplace. If you do, and offend someone, well, you were informed. We try to make everyone feel as comfortable as possible, so that they can focus on their work and studies. Management will crack down very hard if there is reason to fear that people are put into a mental state which compromises safety, on the people who cause such needless grief.
3. Other than that, we are pretty informal in the workplace.
4. SAFETY is the overriding priority. Of course, the most perfect safety is supposed to be achieved when one does absolutely no work and does not even breathe. Our challenge is to devise and observe safety practices while enabling us to get work done on pretty difficult problems. This requires a lot of thought and firm discipline.
5. IF THINGS GO TERRIBLY WRONG, SHUT DOWN ANY MACHINERY ETC, IF YOU CAN DO SO SAFELY.
6. CALL FOR HELP IMMEDIATELY. AND BE ABSOLUTELY CLEAR AND HONEST. This is an easy and sensible rule, and saves trouble later.

## 2.3 EMERGENCY NUMBERS: MEDICAL

Remember, the most important thing is to get immediate aid for the injured!!!

1. Is the injury life-threatening or very serious? If yes-

## CHAPTER 2. GENERAL LAB SAFETY AND SECURITY PROCEDURES<sup>4</sup>

- (a) Apply first aid as necessary.
  - (b) Call police/emergency for help in getting emergency personnel to scene quickly: 0480 282 5228
  - (c) Irinjalakkuda Govt. General Hospital is: 0480 282 5213
  - (d) Notify College Office Office Supervisor: 0480 282 5355 or 830 194 5355. Or Principal: 98475 20278
2. Not life-threatening, but requires immediate attention at a hospital? If yes-
  - (a) Apply first aid as necessary.
  - (b) Call (give phone number)
  - (c) Notify College Office Principal: 98475 20278, Office: 0480 282 5355 or 830 194 5355. Or
3. Is the injury minor, requiring just a visit to a physician, rather than a hospital? If yes-
  - (a) Apply first aid as necessary.
  - (b) Refer to list of approved clinics/physicians. (LIST will be added)
  - (c) Go to clinic/physician
4. FIRE/ACCIDENT: No. : 4802820558. Use extinguishers etc. as possible; alert others around as permitted by your own safety, and get out of the area. TRY NOT TO BE HEROIC: Leave that to the professionals.
5. SECURITY: POLICE : No. 0480 282 5228 ;
6. PEOPLE OUT OF CONTROL Principal 98475 20278 if needed.

### 2.4 Tools and workplace discipline

Keep the place clean and orderly. Anything less is a safety threat and general distraction. Put tools and fasteners away in their proper place at the end of the day before you leave. Clean up metal filings, dust, wood shavings etc. promptly.



## 2.5 Team coordination

You are a member of a team. Look out for your colleagues. Be disciplined about observing schedules and appointments. **COMMUNICATE!** Do not get caught in a position where you have to be complaining about the team not coordinating and communicating properly.

## 2.6 Handling external requests/ cleaning / maintenance

- **Cleaning:** The custodians remove items from trash cans placed **OUTSIDE** the doors. Please do move garbage cans out periodically.
- Floor cleaning/waxing is scheduled periodically, to occur after hours. We **ARE** concerned about equipment etc. so we try to get everything out of the way and locked up if possible. Things **MUST** be moved from the floors before such scheduled cleaning.
- **Apart from the above, all tour requests must be cleared with lab director/College office, and we do not in general accommodate requests for impromptu tours. These do pose safety concerns, so it is perfectly OK to refuse unscheduled tours.**

### Purchasing / Accounts system

- Please contact Prof.Sudha Balagopalan (Assisted by the Management's nominee).
- Purchases must be related (at least make sense) as part of the project from which they are made. We have generally more leeway than industry etc., in that items purchased for one project can be **USED** for another project as well, but the purchase itself must make sense for the project where it is purchases.
- Every purchase request **MUST** be entered in the proper category in the proper account. From time to time, accounts must be reconciled with the Finance Office.



## 2.7 Working with the Workshop: CAD; intelligent design; off-the-shelf vs. part fabrication; scheduling; priorities; charge numbers.

The people in the workshop are our colleagues, who provide expert assistance. However, they are not expected to do your engineering design, thinking or purchasing of items for you: those are your tasks. Note:

- **THEY** decide who is allowed to handle machinery inside the Machine Shop.
- If you borrow tools from them, **CLEAN AND RETURN TOOLS IMMEDIATELY WHEN YOU STOP WORKING. DO NOT LEAVE THEM LYING AROUND ON THE THEORY THAT YOU WILL BE BACK TOMORROW!**
- Generally, machine shop work requires a project charge number. Please contact lab director/ Prof SB if you do not already know under what project to charge specific items.
- Do not specify fabrication from scratch for items which can be substituted by inexpensive, off-the-shelf items with some ingenious design and careful analysis.
- **Engineer:** The term arises from Ingenieur: One who is Ingenious.

## Chapter 3

# Specifics Related to UAVs

1. Whatever the weight class we observe **SAFETY RULES**. All have propellers that spin very fast.
2. **NO ONE STAYS IN OR AHEAD OF THE PROPELLER PLANE** (the circular disc traced by a propeller) while it is operated. Operator shall ensure that everyone is cleared before starting, and should shut down if someone strays there during operations. We cannot predict when a propeller might fail, but pieces generally fly radially out, and forward.
3. **EVERYONE** wears eye protection. No exceptions.
4. Make sure that no one needs to approach the propeller plane from behind either: severe danger.
5. **LiPO** batteries can catch fire. Learn the cautions related to batteries and observe them. Don't try to open LiPo packages. Be aware of risk of shorting them. Damaged LiPos should not be used further until careful examination.
6. **DO NOT FLY UAVs OVER PEOPLE**. **DO NOT** fly near windows. **DO NOT** fly near power lines/cables. Try to stay away from trees (for the UAV's survival).
7. Always have a **WRITTEN CHECKLIST** and **FLIGHT PLAN**, and call out each step so that someone else can confirm that the item is done (Pilot procedure).
8. Check the vehicle before and after flight, clean up any dirt, repair any damage before approving next flight.

9. **DO NOT FLY OVER** restricted areas as given by latest updated NOTAMS and area maps, available on the 'NET. (This is for long after our initial demo tests!!)
10. **DO NOT** fly above 50 feet without further planning, Winds can blow your vehicle off course, you are still responsible.
11. **General aviation rule: PILOT IN COMMAND IS RESPONSIBLE.** Has to decide when and whether to start flight, and when to end it. Decides who specifically operates the vehicle (e.g. student pilot can operate controls, but PIC is responsible). So if PIC is not happy, do not start flight. Arguments can be later.

## Chapter 4

# Chemical Safety

- You are expected to read and think about chemical safety before using any chemicals.
- Protective Equipment is provided. Use it.
- Wear appropriate safety protection gloves and glasses are expected. Wear a mask if there is a risk of exposure to fumes or dust/powder. Get ventilation if needed.

### 4.1 MSDS: Material Safety Data Sheets

An important element described in the law and common sense for chemical safety awareness is the Material Safety Data Sheet (MSDS). Like the label, the MSDS provides you with information regarding the hazards associated with a chemical. However, the information provided is much more comprehensive. It is important to consult an MSDS before introducing a new chemical to your work area or when questions arise while working with hazardous substances. Prepared by its manufacturer, an MSDS provides information to help you understand the intrinsic hazards of the chemical including:

- physical and chemical properties
- stability and reactivity information
- health hazard information

- acute and chronic effects of exposure
- permissible exposure limits MSDS should accompany all products covered under RTK law. If an MSDS is not received, please contact your institution's Lab Coordinator. The Coordinator should be able to obtain the information directly from the manufacture or distributor.

## 4.2 NFPA Hazard Ranking System

The NFPA hazard ranking system shows the type and the degree of a chemical hazard. The labels are diamond-shaped and colour-coded.

- Blue indicates the health hazard.
- Red indicates the fire hazard.
- Yellow indicates the reactivity hazard.
- White gives special information such as water or oxidiser incompatibility.

In each field, the degree of the hazard is rated from 0 to 4, with 4 being the greatest hazard and 0 indicating no significant hazard.

Hazardous Materials Container Labels: If a container is labeled with any of the following terms, it contains a hazardous chemical:

- Corrosives: Destroy living tissue on contact
- Toxics: Hazardous to your health
- Flammables: Readily catch fire
- Reactives: React violently with materials in otherwise stable situations.

## 4.3 Procedure: Chemical Spills

(Ref: Michigan Tech Institute Safety Manual)

Chemical spills that cannot be handled safely by persons working in the area.

### 4.3.1 Chemical Spills Indoors

- A. If the spill represents a hazard that is **Immediately Dangerous to Life or Health, IDLH**, evacuate the room and, if necessary or in doubt, evacuate the building.
- B. Dial (Phone?) to notify office supervisor and request assistance. Describe the situation and whether there is a potential for fire or if there are injuries. Public Safety will contact the Occupational Safety and Health Services emergency spill response coordinator (We will inform the Chemistry department personnel) for assistance. Be prepared to provide a description of the substance spilled and a material safety data sheet, if possible. Also call College office and alert them on the event.
- C. The area of the spill must be secured to prevent students or employees from entering the danger zone.
- D. From a safe location, provide directions to the location and information about the spill to Public Safety or other emergency response personnel as needed.

### 4.3.2 Chemical Spills Outdoors

- A. Dial (Phone) to notify police.
- B. Follow your Department emergency spill response procedures to contain the spill. Do not attempt a cleanup once you have determined that outside assistance is needed, or if the spill has entered the soil, groundwater, or surface water.
- C. Remain a safe distance away from the spill and warn passers-by to stay clear until help arrives. Be prepared to provide information about the substance spilled, including the material safety data sheet, to emergency responders.

## 4.4 Eye Protection

(From the Michigan Tech Safety Manual, page 13) Eye and Face Protection Each affected person shall use appropriate eye or face protection if a hazard exists due to any of the following:

- Flying objects or particles.
- Moving or dangling objects like slings and chains.
- Dusts and mists.

- Molten metal.
- Liquid chemicals.
- Acids or caustic liquids.
- Chemical gases or vapours.
- Glare.
- Injurious radiation.
- Electrical flash.
- Any combination of the above hazards.

Safety requirements for eye protection: Side protection shall be used whenever there is a hazard from flying objects. Spectacles without side shields are allowable for frontal protection only (it should be noted that this situation would be extremely unlikely).

A face or eye protector shall be in compliance with all of the following minimum requirements:

- It shall protect against the particular hazards for which it is designed.
- It shall fit snugly and shall not unduly interfere with movements of the wearer.
- It shall be capable of withstanding sanitising.
- Care shall be taken to recognise the possibility of multiple and simultaneous exposure to a variety of hazards.

## 4.5 Wearing Gloves

Gloves are required when there is a risk of: (source: Michigan Tech Safety Manual)

- Skin absorption of harmful substances.
- Severe cuts or lacerations.
- Severe abrasions.
- Punctures.



- Chemical burns.
- Thermal burns.
- Harmful temperature extremes.

NOTE: MASK and glasses Required when working with things that generate dust: filing, sawing, drilling carbon fiber, metals.

## 4.6 Working Alone

We strongly discourage (i.e., prohibit unless there is some clear overpowering reason and adequate other approved safeguards in place) working alone in the lab at night and on weekends. During normal working hours, if the work involves exposure to hazards, you must not work alone.

From the Michigan Tech Safety Manual:

' hazards that are potentially life threatening, could inhibit self-rescue, could cause injuries requiring immediate assistance, or pose a fire or explosion hazard beyond the person's ability to respond effectively. Appropriate methods to address the need to perform such hazardous operations include the buddy system, intercom communication to a nearby area, periodic supervisor inspections, periodic phone contacts, etc., as long as the method implemented is appropriate to the level of hazard and the required response time in the event of an incident. Each department is responsible for establishing a system and criteria for approving requests to work alone.'

Smart thing is to wait until there can be another person present, who knows whom to call for help, and how to find and operate the phone.

## 4.7 GAS CYLINDERS

From Michigan Tech Safety Manual, pages 21-24:

Flammable Liquids Outside of Laboratories Storage of flammable liquids in buildings shall be limited to that required for the operation of office equipment, maintenance, and classroom demonstrations in addition to the following restrictions:

- Containers of Class IA liquids shall not exceed one pint capacity for glass containers, one gallon capacity for metal, or two gallons capacity for safety cans.
- Containers of Class IB liquids shall not exceed one quart capacity for glass containers, one gallon capacity for metal, or two gallons capacity for safety cans.
- Containers of Class IC liquids shall not exceed one gallon capacity for glass containers, one gallon capacity for metal, or two gallons capacity for safety cans.
- Containers of combustible liquids shall not exceed one gallon capacity for glass containers, or five gallons capacity for metal containers and safety cans.
- Not more than ten gallons (or 25 gallons in safety cans) of Class I and combustible liquids combined shall be stored in a single fire area outside of a UL listed or FM approved storage cabinet or an inside flammable liquids storage area.

#### Definitions:

- Class IA flammable liquid - a liquid having a flash point below 73 F (22.8 C) and having a boiling point below 100 F (37.7 C).
- Class IB flammable liquid - a liquid having a flash point below 73 F (22.8 C) and having a boiling point at or above 100 F (37.7 C).
- Class IC flammable liquid - a liquid having a flash point at or above 73 F (22.8 C) and below 100 F (37.7 C).
- Combustible liquid - a liquid having a flash point at or above 100 F.

#### Compressed Gases

- Compressed gas cylinders, either empty or full, shall be used, handled, and stored in accordance with the following:
- Compressed gases cylinders must be stored in a vertical position and may not be stored in hallways, stairwells, receiving areas, locations where they are subject to damage.
- A chain, bracket, clamp, or other restraining device shall be used at all times to prevent cylinders from falling.

- Acetylene or liquified gas cylinders shall not be placed on their sides, but shall be stood valve-end up.
- A cylinder, whether empty or full, shall not be used as a roller or as a support.
- A cylinder, whether empty or full, in storage or during shipment, shall have the valve closed and cap connected in place, if a cap is provided in the design, or the valve shall be otherwise protected.
- Cylinders shall be marked with either the chemical or trade name. Marking shall be by stencilling, stamping, or labelling and shall not be tampered with or be readily removable. If the labelling is unclear or defaced, return the cylinder or obtain a new label from the supplier. Unlabelled cylinders shall not be used.
- Empty cylinders shall be marked ØemptyÓ or ØMTÓ at time of depletion.
- Cylinders of oxidisers such as oxygen shall be stored at least 20 feet from fuel gas cylinders or a highly combustible material such as, but not limited to, oil, grease, flammable gas or a source of ignition, or be separated from the material by a noncombustible wall, not less than 5 feet high, having a fire resistance rating of one hour. All cylinders shall be stored away from heat in excess of 125 degrees Fahrenheit.
- Where different gases are stored, they shall be grouped by types. Groupings shall separate the flammable gases from the oxidising gases.
- Storage shall be set up to ensure "first-in, first-out" usage.
- A cylinder storage area shall be posted with the names of the individual gases stocked, and a warning posted against tampering by an unauthorised employee. An assigned storage area shall be located where a cylinder will not be knocked over or struck by a passing or falling object.
- A storage area for cylinders shall be well ventilated. A cylinder shall not be stored in basements or pits except where appropriate ventilation is furnished to keep the area purged of any accumulation of gases.
- Cylinders shall be transported in an upright position and securely fastened by a restraining device to the truck or handcart. Approved handcarts are to be used when transporting cylinders within a building.

- When transported, the regulator must be removed and the protective cap replaced.
- A cylinder shall not be dropped, dragged, rolled on its side, or struck violently.
- A cylinder shall be lifted only by enclosed platforms when using a crane or hoisting device. Electromagnets, ropes, or slings shall not be used.
- When transporting cylinders in an elevator, other passengers should not be allowed to occupy the elevator.
- Use cylinders in an upright position and secure them firmly with chains or clamps.
- Do not use a cylinder of compressed gas without reducing the pressure through a regulator attached to the cylinder valve.
- Use regulators and pressure gauges only with gas for which they were designed and intended. Do not use adapters or modify connectors to circumvent this rule.
- Make sure the threads on a regulator or union correspond with those on the cylinder valve outlet. Do not force mismatched connections.
- Never use oil or grease on valves or attachments for oxygen cylinders. Avoid handling oxygen cylinders and apparatus with oily hands, gloves, or clothing.
- Open cylinder valves slowly with valve outlet directed away from personnel. Close the main cylinder valve as soon as it is no longer necessary to have it open.
- Gases shall not be mixed within a cylinder except by the supplier.
- A cylinder shall not be placed where it will become a part of the electrical circuit by accidental grounding or where it may be burned by electric Welding arc. A cylinder shall not be placed so that hot slag or flame will reach it or it shall be protected by a fire resistant shield. An electrode shall not be tapped against a cylinder to strike an arc.
- A frozen or ice-clogged valve shall be thawed either by warm air or use of warm water and dried before using. Boiling water or a flame shall not be used. Force shall not be applied to a valve or cap to loosen a cylinder frozen in place.



- A cylinder without fixed hand wheels shall have keys, handles, or non-adjustable wrenches on valve stems while in service. A multiple cylinder installation shall require only 1 key or handle for each manifold. A hammer shall not be used to open a cylinder valve or loosen a cap.
- A leaking cylinder or a cylinder with a valve stuck open or a valve in need of repair shall be taken outdoors - if it is safe to do so - away from sources of ignition, slowly emptied, tagged with a warning sign, and the manufacturer or distributor notified.
- Complete removal of the stem from the cylinder valve shall be avoided.
- Nothing shall be placed on top of a cylinder which would damage a safety device or interfere with the quick closing of the valve.
- Return empty cylinders to the vendor as soon as possible.Ó

## 4.8 Machine tool safety

(from the GITAE Machine Shop's Procedures)

- No one shall loiter in the machine shop or otherwise disturb the shop mechanics from performing their work.
- No member of the faculty, research faculty, staff, or student shall operate machines unless there is another person in the workshop. Same for the lab!!!
- All persons who operate any rotating machine shall not wear a necktie or have on their person any item of clothing, which may become entangled with the machine. Whenever possible, they shall wear a coat which buttons completely to the collar. Long hair must be contained by net or band.
- No person shall operate any machine which produces chips, swarf, etc., without wearing protective glasses.
- No person shall use a drill press unless the items to be drilled are held in an appropriate clamp.
- No material may be used from stock supplies until an approval has been received from the shop engineer.

- No machine or piece of equipment shall be left dirty. All machines shall be cleaned by the operator at the completion of his/her work or at the end of each working day though an unfinished job remains in the machine.
- No machine is to be left with unfinished work in it for prolonged periods unless absolutely necessary.
- No machine is to be left unattended while it is operating.
- All malfunctions of the equipment and all broken tools shall be reported promptly to the shop engineer or a shop mechanic.
- All tools checked out of the bins shall be cleaned and returned at the end of each working day.
- All accidents must be reported promptly to the AE Administration Office.
- No faculty member, research faculty, staff, or student shall operate any piece of shop equipment until he/she has been cleared to do so by the shop engineer and has signed a statement indicating that he/she has received instructions and fully understands the operation of the equipment and the possible hazards involved.

## 4.9 BIBLIOGRAPHY

- American National Standards Institute, American National Standard for the Safe Use of Lasers: ANSI Z 136.1 (1993), Laser Institute of America, New York, NY (1993).
- American National Standards Institute, American National Standard for the Safe Use of Optical Fiber Communication Systems Utilizing Laser Diode and LED Sources: ANSI Z 136.2 (1988), Laser Institute of America, Orlando, Florida, 1988.
- American National Standards Institute, American National Standard for the Safe Use Lasers in the Health Care Environment: ANSI Z 136.3 (1988), Laser Institute of America, Orlando, Florida, 1988.
- Food and Drug Administration: Regulations for the Administration and Enforcement of the Radiation Control for Health and Safety Act of 1968 (Lasers), Center for Devices and Radiological Health, Food and Drug Administration (DHHS), 21 CFR Chapter 1, 1968.

- Food and Drug Administration: Performance Standard for Laser Products, Center for Devices and Radiological Health, Food and Drug Administration (DHHS), 50 CFR 161: 33682-33702, 1985.
- Food and Drug Administration: Compliance Guide for Laser Products, Center for Devices and Radiological Health, Food and Drug Administration (DHHS), FDA 86-8260.
- Doyle, Daryl and Kokasa, John, Laser Processing of Kevlar: Hazardous Chemical By-products, Proceedings of ICALEO, Laser Institute of America, Toledo, Ohio, 1986.
- Rockwell, R. James, Jr., Ed., Laser Safety Training Manual - Sixth Edition, Rockwell Associates, Inc., Cincinnati, Ohio, 1989.
- Sliney, David H. and Wolbarsh, Myron L., Safety With Lasers and Other Optical Sources. New York, Plenum, 1980.
- Rockwell, R. James, Jr., "Ensuring safety in laser robotics," Lasers and Applications, 3(11): 65-69, November 1984.
- Rockwell, R. James, Jr., "Fundamentals of Industrial Laser Safety." In Industrial Laser Annual Handbook, M. Levitt and D. Belforte, Eds. Penn Well Books, Tulsa, Okla., pp. 131-148, 1986.
- Laser Institute of America, "Safety information on electrical hazards," Laser News, Laser Inst. of America, 6(5): 8-14, September 1984.
- Doyle, D.J. and Kokosa, J.M., "Hazardous By-Products of Plastics Processing with Carbon Dioxide Lasers." In: Laser Welding, Machining and Materials Processing: C. Albright, Ed. Proceedings of ICALEO, IFS LTD., Bedford, U.K., pp. 201-203, 1985.
- Rockwell, R. James, Jr. and Moss, C.E., "Optical radiation hazards of laser welding processes, part I: Nd:YAG laser," The Journal of The American Industrial Hygiene Association, 44(8): 572-579, August 1983.
- Rockwell, R. James, Jr. and Moss, C.E., "Optical radiation hazards of laser welding processes, part II: carbon dioxide laser," The Journal of The American Industrial Hygiene Association, 50(8): 419-427, August 1989.



- Rockwell, R. James, Jr., "Laser accidents: are they all reported and what can be learned from them?" Journal of Laser Applications, Laser Institute of America, Toledo, Ohio, pp. 53-57, October 1989.
- Underwriters Laboratories Inc., UL 1740: Standard for Industrial Robots and Robotic Equipment. Underwriters Laboratories Inc. Northbrook, Ill., 1995.



**Principal**  
**Christ College of Engineering**  
**(Autonomous)**  
**Irinjalakuda - 680 129**